

THEORY OF CHEMICALLY ACTIVATED UNIMOLECULAR REACTIONS. WEAK COLLISIONS AND STEADY STATES

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The prediction of decomposition/stabilization ratios for chemically activated unimolecular reactions is usually based on the assumption of steady state and irreversible deactivation. The levels below the activation energy E_0 are treated as a sink. In the present work we assume that the reaction can be described by a weak-collision master equation. An exact analysis shows that one can expect three timescales: an initial transient, an intermediate steady state, and an asymptotic steady state. The intermediate steady state is well defined only if the eigenvalues of the corresponding thermal rate matrix can be separated into two groups one of which contains eigenvalues of substantially smaller magnitude. An efficient and accurate method for the estimation of intermediate steady-state populations and branching ratios is described. It avoids the irreversible deactivation assumption which is shown to lead to errors particularly for weak collisions. The new analysis is illustrated in a series of calculations for a model of the sec-butyl radical system. The numerical solutions are readily obtained by use of a recently developed equilibrium finite-basis-set method.

1. Introduction

The progress of a unimolecular reaction is commonly described by a master equation obtained from RRKM theory [1,2] which uses a strong collision assumption, i.e. each collision is taken to leave the reactant molecule in a canonical equilibrium distribution. In recent years the failures of the strong-collision assumption have been amply documented [3,4] and a more realistic master equation incorporating a weak-collision energy transfer mechanism has come into use [5–10]. The calculation of a rate coefficient for the thermally activated reaction is based on a steady-state assumption better described as an assumption of exponential decay. The strong-collision assumption allows the numerical work to be reduced to one-dimensional

integration [11] but the calculation is non-trivial in the general weak-collision case where the master equation must be discretized and diagonalized by numerical means. Significant improvements in the computational implementation of the weak-collision theory have recently been described [9,10,12, 13] allowing the effects of energy transfer efficiency to be explored in detail for thermally activated unimolecular reactions.

Weak collisions are prevalent also for chemically activated reactions and their consequences need investigation. One aim of the present work is to bring the efficiency of the recently developed equilibrium finite-basis-set method [13] to bear upon this problem. However, while the basic mechanism of the chemically activated reaction is understood [14,15] the difficulty of the problem has forced the use of a simplified model [16–22] which is not true to the underlying mechanism. Thus we shall restate the problem below and offer

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a new more rigorous analysis of the chemically activated reaction.

In the case of chemically activated unimolecular reactions the attention is generally focused on the relative rates of decomposition and stabilization. One assumes that deactivation as well as reaction are irreversible steps in the reaction mechanism. The flux of reactant molecules created by chemical activation is taken to be perfectly balanced by the decomposition and stabilization fluxes so that the population of energized ($E > E_0$) reactant molecules remains in a steady state. Both types of irreversibility, decomposition or stabilization, are potential sources of error. The former is part of the standard thermal reaction theory as well and we choose to accept it for the time being. The stabilization irreversibility, however, is contrary to the model mechanism of the thermal reaction theory. We shall therefore abandon it in order to achieve a more consistent unified theory of unimolecular reactions. In this theory the population of reactant molecules at all levels will be explicitly included and the contribution of collisional energization considered. This forces us to reconsider the meaning of the usual steady-state assumption and to investigate the conditions under which it will exist.

Our basic theory is described in section 2 and compared to the usual irreversible stabilization ("sink") theory. An exact analysis of the consequences of our theory for the initial, intermediate and asymptotic behavior of the reactant population and branching ratios is given in section 3. The conditions for the existence of a well-defined intermediate steady state are obtained and approximate expressions for the corresponding level populations and branching ratios are derived. The calculations for the sec-butyl radical reaction carried out by the equilibrium finite-basis-set method as an illustration of the content of our theory are described in section 4. The results are discussed in section 5 with concluding comments in section 6.

2. Basic theory

2.1. The chemical activation master equation

The master equation describing the time evolu-

tion of the reactant molecule probability density $p(t; E)$ for a chemical activation system can be written as [1,2,14],

$$\partial p(t; E) / \partial t = \omega \int_0^\infty dE' P(E, E') p(t; E') - [\omega + k_a(E)] p(t; E) + f(t; E), \quad (1)$$

using the notation of ref. [13]. Compared to the analogous equation for thermal unimolecular reactions one has an additional term, $f(t; E)$, accounting for the incoming flux of activated reactant molecules. This flux is assumed to be into an energy region beginning at an energy E_F which is above the critical energy for unimolecular reaction E_0 so that the total incoming flux is,

$$F(t) = \int_{E_F}^\infty dE f(t; E). \quad (2)$$

In the usual treatments one assumes that these activated molecules may either decompose unimolecularly with a total flux rate $D(t)$ given by,

$$D(t) = \int_{E_0}^\infty dE k_a(E) p(t; E), \quad (3)$$

or that they may be stabilized with a flux rate $S(t)$. Thus, we have the conservation condition

$$F(t) = D(t) + S(t). \quad (4)$$

Integration of eq. (1) over E and employing the normalization of $P(E, E')$ yields,

$$\int_0^\infty dE \partial p(t; E) / \partial t = - \int_{E_0}^\infty dE k_a(E) p(t; E) + \int_{E_F}^\infty dE f(t; E),$$

or by (2)–(4)

$$S(t) = \int_0^\infty dE \partial p(t; E) / \partial t. \quad (5)$$

Complaints may be raised that this definition takes molecules at $E > E_0$, about to decay, to be stabilized. The traditional sink method lends itself to the interpretation that only molecules of energy less than some chosen lower cutoff are stabilized. While this may seem physically more plausible the fact is that no molecules will ever be "stabilized".

The collisional deactivation mechanism is always accompanied by a corresponding activation mechanism which makes any division of reactant molecules into stabilized and unstabilized forms arbitrary. Thus we choose here to refer to all reactant molecules as stabilized. This definition enjoys the advantage of simplicity and precision. The rate of stabilization so obtained will nearly always be to within relevant accuracy identical to that obtained by considering only molecules below E_0 as stabilized.

2.2. The traditional steady-state sink method

The assumption of steady state would normally be taken to mean that the population is time independent,

$$\partial p(t; E)/\partial t = 0, \quad 0 < E < \infty. \quad (6)$$

Such an assumption is, in fact, justified as we shall see below, if the decay to products is irreversible and t is large enough. In this case there is perfect balance between incoming flux and thermal or direct reaction and the stabilization rate vanishes. However, the branching ratio D/S is often observed to take on a stable finite value on the relevant experimental timescale. This is understood to be the consequence of an effectively irreversible de-energization of reactant molecules. This amounts to an assumption that any activation by collisional means can be neglected, i.e.

$$\int_0^{E_0} dE' P(E, E') p(t; E') \approx 0, \quad E > E_0. \quad (7)$$

The reactant energy levels below E_0 are treated as a probability "sink".

Thus, using eqs. (1) and (7), the traditional sink formulation treats levels above E_0 via the modified master equation,

$$\begin{aligned} \partial p(t; E)/\partial t = & \omega \int_{E_0}^{\infty} dE' P(E, E') p(t; E') \\ & - [\omega + k_a(E)] p(t; E) + f(t; E), \\ & E > E_0. \end{aligned} \quad (8)$$

The steady-state assumption is modified to,

$$\partial p(t; E)/\partial t = 0, \quad E > E_0, \quad (9)$$

and combining (8) and (9) yields,

$$\begin{aligned} & [\omega + k_a(E)] p(t; E) \\ & - \omega \int_{E_0}^{\infty} dE' P(E, E') p(t; E') \\ & = f(t; E), \quad E > E_0. \end{aligned} \quad (10)$$

Symmetrizing this equation as in ref. [13] yields,

$$\omega \int_{E_0}^{\infty} dE' S(E, E') q(t; E') = \hat{f}(t; E), \quad E > E_0, \quad (11)$$

where,

$$\begin{aligned} S(E, E') = & [k_a(E)/\omega + 1] \delta(E - E') \\ & - P(E, E') r(E')/r(E), \end{aligned} \quad (12)$$

$$q(t; E) = p(t; E)/r(E), \quad (13)$$

$$\hat{f}(t; E) = f(t; E)/r(E), \quad (14)$$

$$r(E) = [p_T(E)]^{1/2}, \quad (15)$$

where $p_T(E)$ is the canonical probability density at a temperature T [11].

Either using a finite difference approach [9] or a finite-basis-set method [13] eq. (11) can be expressed in matrix form as,

$$\omega \mathbf{S} \mathbf{q}' = \mathbf{f}', \quad (16)$$

where the prime indicates that the first row and column are missing from all matrices [23]. A formal solution of (16) is,

$$\mathbf{q}' = \omega^{-1} (\mathbf{S}')^{-1} \mathbf{f}'. \quad (17)$$

3. An exact analysis

3.1. General solution

The chemically activated master equation in (1) can be solved exactly in a formal sense. Using the symmetrizing transformations (12)–(15) we can write the master equation in the form,

$$\begin{aligned} \partial q(t; E)/\partial t = & -\omega \int_0^{\infty} dE' S(E, E') q(t; E') \\ & + \hat{f}(t; E), \end{aligned} \quad (18)$$

or equivalently in Dirac notation as,

$$\partial|q(t)\rangle/\partial t = -\omega \mathbf{S}|q(t)\rangle + |\hat{f}(t)\rangle. \quad (19)$$

Eq. (19) can be formally solved by the integrating-factor method as,

$$|q(t)\rangle = \exp(-\omega \mathbf{S}t)|q(0)\rangle + \int_0^t ds \exp[-\omega \mathbf{S}(t-s)]|\hat{f}(s)\rangle. \quad (20)$$

Supposing the eigenvalues and eigenvectors of $\mathbf{S}$ are known in a N -dimensional basis,

$$\mathbf{S}|\phi_l\rangle = \lambda_l|\phi_l\rangle, \quad l=1, \dots, N, \quad (21)$$

then we can expand,

$$|q(t)\rangle = \sum_{l=1}^N q_l(t)|\phi_l\rangle, \quad (22)$$

$$|\hat{f}(t)\rangle = \sum_{l=1}^N f_l(t)|\phi_l\rangle. \quad (23)$$

If the eigenvectors are orthonormal,

$$\langle\phi_k|\phi_l\rangle = \delta_{kl}, \quad (24)$$

then,

$$q_k(t) = \langle\phi_k|q(t)\rangle. \quad (25)$$

$$f_k(t) = \langle\phi_k|\hat{f}(t)\rangle. \quad (26)$$

Projecting the eigenvectors of $\mathbf{S}$ on to eq. (20) and applying eqs. (21)–(26) yields

$$q_l(t) = \exp(-\omega\lambda_l t)q_l(0) + \int_0^t ds \exp[-\omega\lambda_l(t-s)]f_l(s). \quad (27)$$

As with usual treatments we will assume the chemically activated input flux is stationary in time,

$$f(t; E) = f(E), \quad (28)$$

and that there is no initial population of reactant molecules,

$$q_l(0) = 0, \quad l=1, \dots, N. \quad (29)$$

Then eq. (27) simplifies to,

$$q_l(t) = (f_l/\omega\lambda_l)[1 - \exp(-\omega\lambda_l t)]. \quad (30)$$

One can then derive exact expressions for the corresponding decomposition and stabilization

fluxes. The decomposition flux is given by (3),

$$D(t) = \int_{E_0}^{\infty} dE k_a(E) r(E) q(t; E) = \sum_{l=1}^N (f_l/\omega\lambda_l)[1 - \exp(-\omega\lambda_l t)] \times \int_{E_0}^{\infty} dE k_a(E) r(E) \phi_l(E). \quad (31)$$

Now noting from (12) and (21) that,

$$\lambda_l = \langle r|\mathbf{S}|\phi_l\rangle/\langle r|\phi_l\rangle = \omega^{-1} \int_{E_0}^{\infty} dE k_a(E) r(E) \phi_l(E) \times \left(\int_{E_0}^{\infty} dE r(E) \phi_l(E) \right)^{-1} \quad (32)$$

and defining,

$$r_l = \langle r|\phi_l\rangle, \quad (33)$$

then,

$$D(t) = \sum_{l=1}^N r_l f_l [1 - \exp(-\omega\lambda_l t)]. \quad (34)$$

Similarly the stabilization flux defined by (5) is,

$$S(t) = \int_0^{\infty} dE \partial p(t; E)/\partial t = (\partial/\partial t) \langle r|q(t)\rangle = \sum_{l=1}^N r_l \partial q_l(t)/\partial t = \sum_{l=1}^N r_l f_l \exp(-\omega\lambda_l t). \quad (35)$$

Note that the conservation-law equation (4) is obeyed since from (34) and (35),

$$S(t) + D(t) = \sum_{l=1}^N r_l f_l = \sum_{l=1}^N \langle r|\phi_l\rangle \langle \phi_l|\hat{f}\rangle = \langle r|\hat{f}\rangle = \int_{E_F}^{\infty} dE f(E) = F.$$

Before relating the exact method to the traditional steady-state method we shall first consider some properties of the exact solution.

3.2. The incubation period

Consider a time much shorter than the time-scale of eigenvalues with appreciable overlap on

$|\hat{f}\rangle$ and $|r\rangle$, i.e. assume that the flux overlap satisfies

$$r_l f_l = \langle r | \phi_l \rangle \langle \phi_l | \hat{f} \rangle \approx 0, \quad (36)$$

for some $l > L$ where $L \leq N$. Then given a time-scale,

$$t \ll 1/\omega\lambda_L < \dots < 1/\omega\lambda_1 \quad (37)$$

we find from (34) and (35),

$$D(t) \approx \omega t \sum_{l=1}^L r_l f_l \lambda_l, \quad (38)$$

$$S(t) \approx \sum_{l=1}^L r_l f_l, \quad (39)$$

and,

$$\omega D/S \approx \omega^2 t \sum_{l=1}^L r_l f_l \lambda_l \left(\sum_{l=1}^L r_l f_l \right)^{-1} \quad (40)$$

For longer times the decomposition flux increases in a multiply exponential manner as shown in (34) and the stabilization decreases in a corresponding manner as in (35). As t increases the fastest (largest $\omega\lambda_l$) components in the multiply exponential decay "burn out" resulting in a slower rate of change of $D(t)$ and $S(t)$.

3.3. The final steady state

At long times,

$$1/\omega\lambda_N < \dots < 1/\omega\lambda_1 \ll t, \quad (41)$$

eqs. (34) and (35) reduce to,

$$D(t) \approx \sum_{l=1}^N \langle r | \phi_l \rangle \langle \phi_l | \hat{f} \rangle \approx \langle r | \hat{f} \rangle = F, \quad (42)$$

$$S(t) \approx 0. \quad (43)$$

Hence given enough time all the chemically activated flux will decompose via thermal reaction. This, however, is not the timescale usually considered in chemical activation analysis.

3.4. The intermediate steady state

Numerous experimental observations indicate that an intermediate timescale exists where the

D/S branching ratio is a finite constant value for a considerable length of time. It is this intermediate "steady state" which is predicted by the usual theory of chemical activation employing the approximation of irreversible stabilization (deactivation sink). For generality assume one can divide the eigenvalue spectrum according to a time t ,

$$1/\omega\lambda_N < \dots < 1/\omega\lambda_K \ll t \ll 1/\omega\lambda_J < \dots < 1/\omega\lambda_1, \quad \text{for } 1 \leq J < K \leq N, \quad (44)$$

where,

$$r_l f_l \approx 0, \quad J+1 \leq l \leq K-1, \quad (45)$$

i.e. the set $\lambda_{J+1}, \dots, \lambda_{K-1}$ are associated with eigenvectors with negligible flux overlaps.

Given such a separation of timescales then,

$$D(t) \approx \sum_{l=K}^N r_l f_l, \quad (46)$$

$$S(t) = F - D(t) \approx \sum_{l=1}^J r_l f_l. \quad (47)$$

In a lot of cases this intermediate timescale can be specified simply by

$$1/\omega\lambda_2 \ll t \ll 1/\omega\lambda_1, \quad (48)$$

and the decomposition and stabilization fluxes reduce to,

$$D(t) \approx \sum_{l=2}^N r_l f_l, \quad (49)$$

$$S(t) = F - D(t) \approx r_1 f_1. \quad (50)$$

In practical applications the expressions in (47) and (50) are the most convenient to use since one only requires the lowest set of J eigenvectors to yield the stabilization flux with the decomposition flux obtainable via the conservation condition. This compares favourably to the complete matrix inversion or solution of a set of simultaneous equations as required by the traditional sink methods. The use of eqs. (44)–(50) will be termed the intermediate timescale approach.

The corresponding intermediate timescale population vector can be obtained from eqs. (22),

(30), (44) and (45) as,

$$|q(t)\rangle \approx t \sum_{i=1}^J f_i |\phi_i\rangle + \sum_{i=K}^N (f_i/\omega\lambda_i) |\phi_i\rangle \\ = t|q_S\rangle + |q_{SS}\rangle, \quad (51)$$

where $|q_{SS}\rangle$ plays the role of a steady state population. We note that the total reactant population is not in a steady state but the reactant probability grows linearly with time. However, the linear growth occurs on a "stabilized" population vector $|q_S\rangle$ which can be expected to have very little overlap with levels above E_0 where $|q_{SS}\rangle$ gives a good approximation to $|q(t)\rangle$. This is the reason for the success of the traditional sink method. One can readily calculate $|q_{SS}\rangle$ approximately from (19) and (51) as,

$$|q_{SS}\rangle \approx \omega^{-1} \mathbf{S}^{-1} (|\hat{f}\rangle - |q_S\rangle). \quad (52)$$

The advantage of this expansion is that $|q_{SS}\rangle$ is obtained by inversion of the rate matrix but without the need to know more than the lowest J eigenvectors of $\mathbf{S}$.

4. Calculations

In order to compare the sink, the exact and the intermediate timescale methods a number of calculations were carried out on a typical system. It must be stressed that these calculations were per-

Table 1

The average thermal energy of the reactant $\langle E \rangle_T$ (kcal/mol) and the reactant-reactant collision frequency ω (s^{-1}) as a function of temperature T (K) and pressure P (Pa)

T (K)	$\langle E \rangle_T$	P (Pa)	ω (s^{-1})
300	1.64	10^{-3}	1.5×10^2
		10^5	1.5×10^{10}
373	2.68	10^{-3}	1.4×10^2
		10^5	1.4×10^{10}
500	5.26	10^{-3}	1.2×10^2
		10^5	1.2×10^{10}
600	7.89	10^{-3}	1.1×10^2
		10^5	1.1×10^{10}
1200	31.3	10^{-3}	7.6×10^1
		10^5	7.6×10^9

formed to look at differences between the above theories and have not been optimized with respect to grainsize and experimental parameters.

As an example of a typical system the sec-butyl radical system was chosen, being much studied [18–20,24–26]. Calculations were performed at relatively low temperatures, $T = 300$ – 600 K, up to relatively high temperatures, $T = 1200$ K, to examine the breakdown of the steady state and irreversible sink assumptions. Average thermal energies of the reactant $\langle E \rangle_T$ corresponding to such temperatures are given in table 1 in kcal/mol. Collisional models such as the strong-collision assumption (SCA), the exponential (EXP) and the stepladder (SL) were examined [13]. For simplicity a monoenergetic chemical input flux was assumed with an input energy $E_F = 40$ kcal/mol which is above the critical energy for thermal unimolecular reaction, $E_0 = 33$ kcal/mol.

To numerically implement the above methods the equation must be converted to matrix form. Instead of the more usual coarse-grained quantum-level approach [9] the recently developed equilibrium finite-basis-set method was employed [13]. Essentially, all energy-dependent quantities are expanded in terms of a finite basis set made up of consecutive sections of the equilibrium probability density $p_T(E)$. It is better suited to evaluating the lowest eigenvalue and corresponding eigenvector of the thermal-rate problem but allows for flexible graining schemes so that it is easy to adapt to the needs of resolving other energy-dependent quantities such as the chemical flux distribution $f(t; E)$.

All calculations were performed with the same grainsize of 1 kcal/mol and although smaller grainsizes are easily possible such further optimization is unnecessary for present purposes. The highest upper-energy cutoff used was $E_H = 100$ kcal/mol at $T = 1200$ K so that the maximum number of grains and thus the maximum order of the rate matrix $\mathbf{S}$, was $N = 100$.

To help relate the system to experimental conditions the collision frequency ω was obtained from the gas pressure P by assuming binary hard-sphere collisions (see ref. [1], p. 164), i.e.

$$\omega = [8\pi/\mu k_B T]^{1/2} \sigma_d^2 P, \quad (53)$$

where the reduced mass μ is 28.6 g/mol and the

hard-sphere collision diameter σ_d is 6.5 Å for reactant–reactant collisions. A set of collision frequencies so obtained are given in table 1. More accurate methods of obtaining collision frequencies exist [27] but are unnecessary for the present model calculations.

5. Discussion

5.1. The behavior of eigenvalues and flux overlaps

Consider the exact expressions eqs. (34) and (35). One can see that the time development of $D(t)$ and $S(t)$ is crucially related to the eigenvalue spectrum λ_i and the flux overlaps $r_i f_i$. These quantities are temperature and pressure dependent. It is instructive to define a set of lifetimes as,

$$\tau_i = 1/\omega\lambda_i. \quad (54)$$

The physical meaning of such lifetimes may become clearer if one considers the relationship of the eigenvalues with the relaxation processes present, i.e. collisional activation/deactivation and unimolecular decomposition.

One can write the thermal rate matrix $\mathbf{S}$ in EFBS notation [13] as,

$$S_{lm} = (K_m/\omega N_m^2 + 1)\delta_{lm} - I_{lm}/N_l N_m, \quad (55)$$

where for the m th grain or basis function,

$$K_m = \int_{E_{m-1}}^{E_m} dE k_a(E) p_T(E), \quad (56)$$

$$N_m = \left(\int_{E_{m-1}}^{E_m} dE p_T(E) \right)^{1/2}, \quad (57)$$

$$I_{lm} = \int_{E_{l-1}}^{E_l} dE \int_{E_{m-1}}^{E_m} dE' P(E, E') p_T(E'). \quad (58)$$

In this notation K_m/N_m^2 can be thought of as the average microscopic rate constant associated with the m th grain or basis function.

Let us first consider the high-pressure behavior of the chemical activation system at low temperatures. As $\omega \rightarrow \infty$,

$$S_{lm} \rightarrow \delta_{lm} - I_{lm}/N_l N_m. \quad (59)$$

In the case of a strong-collision (SCA) model [13],

$$P(E, E') = p_T(E)$$

and hence by eqs. (57)–(59), as $\omega \rightarrow \infty$,

$$S_{lm} \rightarrow \delta_{lm} - N_l N_m. \quad (60)$$

It can be readily shown that the lowest eigenvalue is near zero with near-degenerate higher eigenvalues of unity above. For a weak-collision model the eigenvalue spectrum is similar but the higher set of eigenvalues are not as degenerate. This behavior can be seen in table 2. The table also shows the corresponding flux overlaps $r_i f_i$. Whether these overlaps are non-zero or negligible is critical to the time development of the fluxes. In the high-pressure limit the collisional deactivation process dominates so that most of the incoming flux F is initially stabilized to S flux. Given times long enough thermal reaction becomes important and all flux eventually becomes D flux. This explains why in table 2 the flux associated with thermal reaction $r_1 f_1 \approx F$ while all the other flux overlaps are small by comparison.

From table 2 one can also see that the strong-collision and weak-collision models differ in the distribution of flux overlap. Under the strong-collision model deactivation of the incoming flux is essentially a single step process with the timescale of the collision frequency as shown by the eigenvalues of unity in table 2a. Thus flux is concentrated at the incoming energy E_F and at stabilized energies which for low temperatures means largely below E_0 . Hence decomposition can only occur from flux at E_F or from stabilized flux activated by thermal reaction. Now, at low temperature the strong-collision rate matrix $\mathbf{S}$ as given by (60) is almost diagonal for the high-energy basis functions since for an energy E_l which is much higher than thermal energies (for example compare E_F with thermal energies in table 1) $p_T(E) \approx 0$ and hence $N_l \approx 0$. Thus the incoming flux at $E_F = 40$ kcal/mol can be associated with an eigenvector $|\phi_F\rangle (= |\phi_{40}\rangle$ in the present set of calculations with a 1 kcal/mol grainsize) which has its support localized to a narrow region around the energy E_F . The stabilized flux on the other hand is associated with $|\phi_1\rangle$ the eigenvector corresponding to thermal reaction. Thus the decomposi-

Table 2

Typical eigenvalues λ_i and flux overlaps $r_i f_i / F$ for the low-temperature ($T = 600$ K) and high-pressure ($P = 10^5$ Pa) region. The lowest eigenvalue λ_1 is related to the thermal rate coefficient [13] while the remaining set of higher eigenvalues are related to the microscopic relaxation processes of collision and decomposition (see section 5.1). For comparison (a) a strong collision model (SCA) and (b) an exponential model (EXP) with $\alpha = 4$ kcal/mol are shown

i	λ_i	$r_i f_i / F$
(a) SCA		
1	3.39×10^{-9}	0.99978
2	1.00	-2.5×10^{-23}
33	1.00	6.6×10^{-25}
34	1.00	3.2×10^{-14}
39	1.00	2.8×10^{-10}
40	1.00	2.2×10^{-4}
41	1.00	-6.8×10^{-10}
(b) EXP		
1	3.35×10^{-9}	0.99892
2	0.223	1.8×10^{-4}
33	0.965	5.1×10^{-6}
34	0.968	4.2×10^{-6}
39	0.978	8.4×10^{-7}
40	0.979	3.3×10^{-6}
41	0.981	3.4×10^{-6}

tion flux (34) can be written for the strong-collision model as,

$$D(t) = r_1 f_1 [1 - \exp(-\omega \lambda_1 t)] + r_F f_F [1 - \exp(-\omega \lambda_F t)]. \quad (61)$$

For weak-collision models on the other hand the deactivation of the flux is a multistep process on several timescales. This is because it may take more than a single collision for deactivation and flux may go to lower or higher decomposing levels than an energy E_F . Thus the situation is more complex than in the strong-collision case and it suffices to note that since more than one timescale is involved in the relaxation process the flux overlap is spread over a wide range of eigenvalues as shown in table 2b. Furthermore, unlike the strong-collision case, $|\phi_F\rangle$ is not localized to a narrow energy range around E_F .

With respect to chemical activation experiments the low-pressure and low-temperature region is of considerable interest in the investigation of both the collisional and the decomposition processes. In

the low-pressure limit, unimolecular decomposition dominates the collisional processes at energies above E_0 . In this limit the diagonal elements for basis functions above the critical energy E_0 are given by,

$$S_{ii} \approx K_i / \omega N_i^2, \quad E_i > E_0. \quad (62)$$

For reasonable collisional models the off-diagonal elements are smaller than these diagonal elements so that the higher eigenvalues are closely associated with the microscopic rate constant,

$$\lambda_i \approx K_i / \omega N_i^2, \quad E_i > E_0. \quad (63)$$

Furthermore, because of this diagonality of $\mathbf{S}$ for energies above E_0 the eigenvectors $|\phi_i\rangle$ corresponding to eigenvalues described by (63) are confined to the narrow energy region $[E_{i-1}, E_i]$ spanned by the i th basis function. Thus, unlike the high-pressure region discussed above, even weak-collision models will have higher eigenvectors localized to narrow energy regions. In particular $|\phi_F\rangle$ (or $|\phi_{40}\rangle$) will be localized to an energy around E_F .

At low pressures, much of the flux has time to decompose before being deactivated so that most of the flux overlap is concentrated on the eigenvector $|\phi_F\rangle$ which is associated with the input energy E_F . As before, any flux that is actually stabilized

Table 3

As for table 2 but for the low-temperature ($T = 600$ K) and low-pressure ($P = 10^{-3}$ Pa) region

i	λ_i	$r_i f_i / F$
(a) SCA		
1	5.70×10^{-5}	4.6×10^{-5}
2	1.00	2.2×10^{-23}
33	1.00	6.2×10^{-26}
34	57.9	9.5×10^{-10}
39	1.17×10^4	2.0×10^{-10}
40	2.18×10^4	0.99995
41	3.81×10^4	-4.7×10^{-11}
(b) EXP		
1	2.42×10^{-5}	6.0×10^{-6}
2	0.237	1.1×10^{-7}
33	1.17	1.8×10^{-33}
34	57.8	1.9×10^{-6}
39	1.17×10^4	1.4×10^{-5}
40	2.18×10^4	0.99997
41	3.81×10^4	-5.5×10^{-6}

Table 4

As for table 2 but for the high-temperature ($T=1200$ K) and high-pressure ($P=10^5$ Pa) region

i	λ_i	$r_i f_i / F$
(a) SCA		
1	8.7×10^{-3}	1.006
2	1.00	-3.2×10^{-17}
33	1.00	2.0×10^{-16}
34	1.00	2.0×10^{-10}
39	1.00	7.8×10^{-6}
40	1.00	3.9×10^{-4}
41	1.00	-4.4×10^{-5}
(b) EXP		
1	3.5×10^{-5}	0.974
2	0.131	8.9×10^{-2}
33	0.993	2.2×10^{-3}
34	0.993	2.3×10^{-3}
39	0.995	-2.0×10^{-10}
40	0.996	-8.1×10^{-14}
41	0.997	-1.5×10^{-19}

will eventually decompose via thermal reaction. Table 3 shows a typical set of eigenvalues and flux overlaps in the low-pressure and low-temperature region.

The differences between strong and weak collision models are accentuated in the low-pressure limit. The SCA model, being essentially a one-step deactivation model, has as the dominant processes

Table 5

As for table 2 but for the high-temperature ($T=1200$ K) and low-pressure ($P=10^{-3}$ Pa) region

i	λ_i	$r_i f_i / F$
(a) SCA		
1	0.408	1.9×10^{-5}
2	1.00	-5.4×10^{-25}
33	1.00	2.2×10^{-24}
34	88.6	1.0×10^{-6}
39	1.7×10^4	1.7×10^{-6}
40	3.1×10^4	0.99998
41	5.5×10^4	-9.2×10^{-7}
(b) EXP		
1	7.4×10^{-2}	2.5×10^{-6}
2	0.427	7.3×10^{-7}
33	4.53	4.9×10^{-12}
34	88.5	1.0×10^{-6}
39	1.7×10^4	7.6×10^{-6}
40	3.1×10^4	0.99999
41	5.5×10^4	-4.4×10^{-6}

either decomposition from an energy E_F or instantaneous stabilization to below E_0 . Hence decomposition can only occur from flux at E_F or from flux activated by thermal reaction. Thus, as in the high-pressure case described above, most of the flux overlap is associated with the eigenvectors $|\phi_1\rangle$ and $|\phi_F\rangle$ and hence the decomposition flux can again be written in the form of eq. (61). On the other hand, weak-collision models can only be approximately represented in this form since they have significant flux overlap associated with many timescales as shown in table 3b.

Most chemical activation experiments are performed at low temperatures and while concentrating on this region it is also useful to briefly consider effects at higher temperatures. A major effect of higher temperatures is that the eigenvalue spectrum becomes more dense. The collisional energy transfer is less dominated by deactivation, as is the case at low temperatures, but starts to show significant effects of corresponding activation. Thus, a larger number of eigenvalues of similar magnitude are important as shown by table 4. A similar narrowing of the spread of eigenvalues occurs at low pressures and high temperatures as can be seen in table 5.

5.2. The existence of an intermediate steady state

Here we concern ourselves with the exact time development and in particular in defining the intermediate steady state. In chemical activation systems one often defines an average or observable chemical activation rate constant $\omega D/S$. In figs. 1 and 2 this quantity has been plotted as a function of time in the low-pressure region ($P=10^{-3}$ Pa). Typical lifetimes τ_i have been noted along the time axis and give some idea of the eigenvalues or relaxation processes important at each time. The behavior of the strong-collision model in fig. 1 is well described by the simple form of eq. (61), derived in section 5.1, which bases the time evolution solely on $|\phi_1\rangle$, the eigenvector corresponding to thermal reaction, and $|\phi_F\rangle$ (or $|\phi_{40}\rangle$), the eigenvector localized in the region of the incoming flux F at $E_F=40$ kcal/mol. The weak-collision model in fig. 2 exhibits similar behavior but because of significant overlap with eigenvectors other than

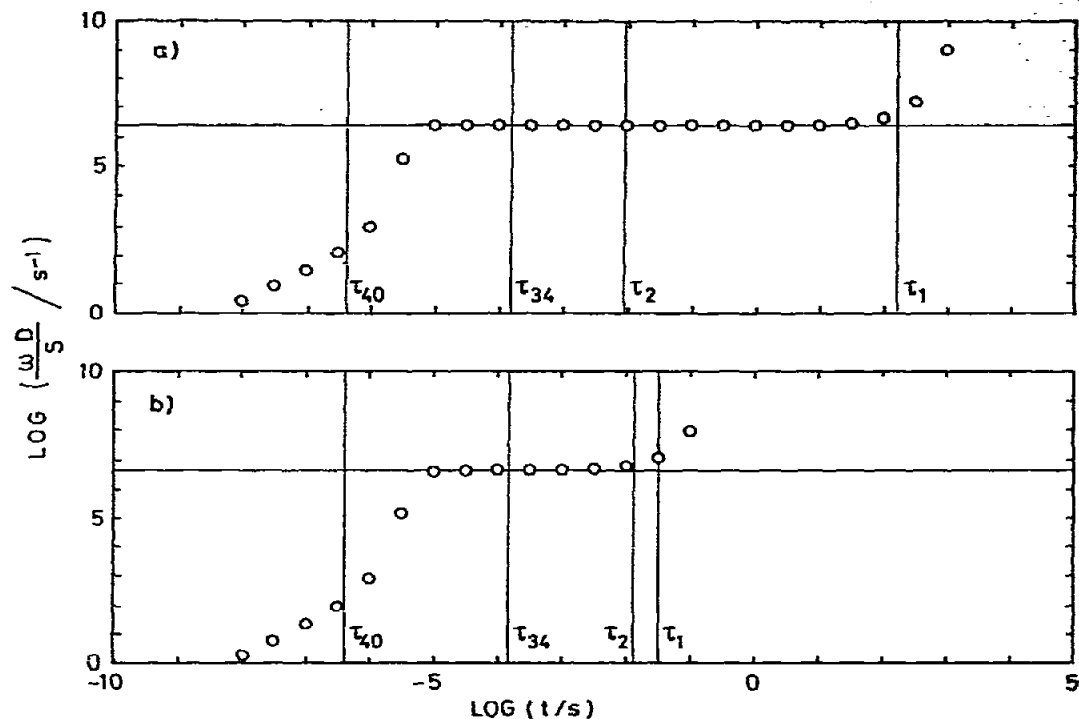


Fig. 1. Chemical activation rate constant $\omega D/S$ as a function of time $t(s)$ for a strong-collision (SCA) model at low pressure ($P = 10^{-3} \text{ Pa}$). (a) The behavior in the low-temperature ($T = 600 \text{ K}$) and (b) the behavior in the high-temperature ($T = 1200 \text{ K}$) region is illustrated for comparison. The points are calculated from the exact equations (34) and (35). The solid horizontal line represents the traditional sink (17) and intermediate timescale results (47) which are indistinguishable on the logarithmic scale of the figure. Tables 9 and 10 illustrate the differences between these two models in a more convenient form. Typical lifetimes $\tau_i(s)$ (54) are given to show the effect of the separation of timescales on the time development.

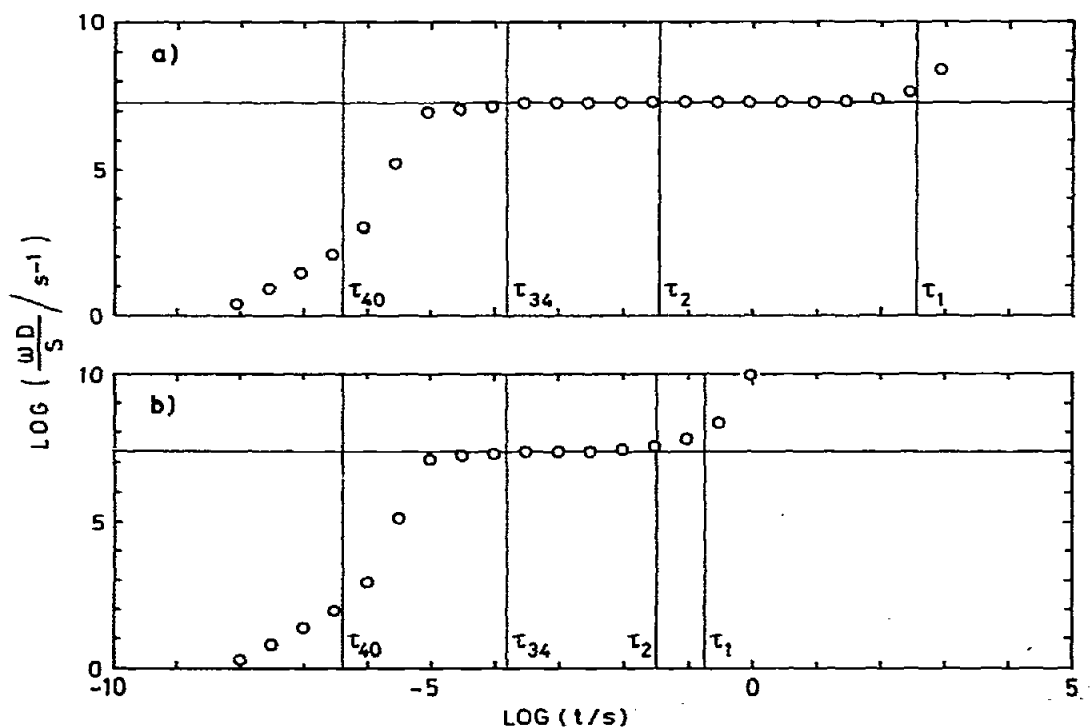


Fig. 2. As for fig. 1, but for an exponential ($\alpha = 4 \text{ kcal/mol}$) collision model at low pressure ($P = 10^{-3} \text{ Pa}$).

Table 6

The length of the intermediate steady state τ (s) as a function of temperature T (K), pressure P (Pa) and collision model (energy jumps α in kcal/mol). This lifetime τ is arbitrarily defined as the time for which there is less than 10% variation in $\omega D/S$ from the intermediate timescale value

Model	T (K)	P (Pa)	$\log \tau$ (s)
SCA	600	10^{-3}	6.0
EXP ($\alpha = 4$)			3.5
SCA	600	10^5	6.0
EXP ($\alpha = 4$)			3.0
SCA	1200	10^{-3}	2.5
EXP ($\alpha = 4$)			≈ 1

$|\phi_1\rangle$ and $|\phi_{40}\rangle$ (see table 3), a more gradual approach to the steady state is apparent. Thus, both models show that, in the low-pressure region, τ_F (τ_{40} in present calculations), the lifetime of the eigenstate localized in the region of the incoming flux F , is the dominant lifetime governing the transition from the incubation period to the intermediate steady state. At times shorter than τ_F the

chemical activation rate constant has behavior typical of the incubation period as given by eq. (40). At times greater than $\approx 10\tau_F$ the rate constant achieves the intermediate-steady-state value. At much longer times, of course, thermal reaction becomes important and the intermediate steady state begins to break down at times longer than $0.1\tau_1$.

As the temperature is raised the eigenvalue spectrum becomes more dense. For example, for the SCA model, $\log(\tau_1/\tau_{40})$ drops from 8.6 to 4.9 as the temperature is raised from 600 to 1200 K. From table 6 one can see that the length of the intermediate steady state suffers a corresponding shortening. At a high enough temperature the intermediate steady state will cease to exist. Instead one will only have an incubation period and a final steady state for times longer than τ_1 .

The disappearance of the intermediate steady state with increase in temperature is more noticeable at higher pressures as shown by fig. 3. At high pressures the higher eigenvalues are all of similar magnitude and an increase in temperature will

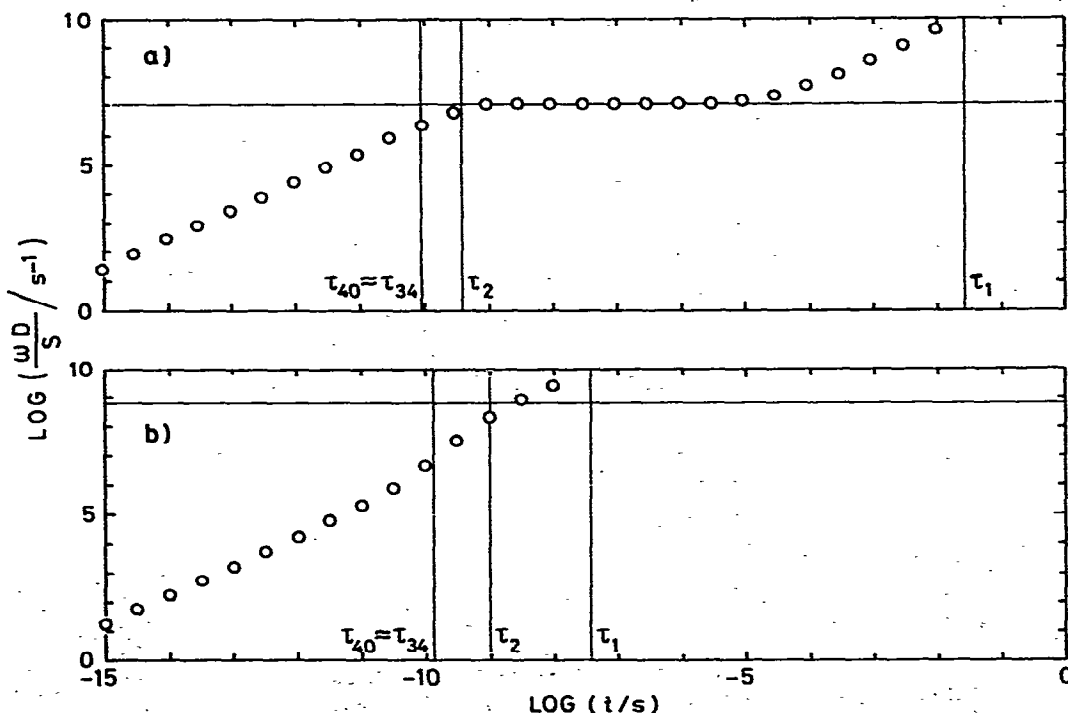


Fig. 3. As for fig. 1, but for an exponential ($\alpha = 4$ kcal/mol) collision model at high pressure ($P = 10^5$ Pa).

Table 7

Application of the intermediate timescale-method equation (47) for an exponential model ($\alpha = 4$ kcal/mol) at various temperatures T (K) and pressures P (Pa)

T (K)	P (Pa)	i	λ_i	$r_i f_i / F$	$(S/F)_i$	$(D/F)_i$	$(\omega D/S)_i$
600	10^{-3}	1	2.42×10^{-5}	6.02×10^{-6}	6.02×10^{-6}	1.000	1.78×10^7
		2	0.237	1.14×10^{-7}			
		3	0.370	2.19×10^{-7}			
		4	0.504	1.81×10^{-7}			
600	10^5	1	3.35×10^{-9}	0.9989	0.9989	1.08×10^{-3}	1.16×10^{-7}
		2	0.223	1.78×10^{-4}			
		3	0.283	1.55×10^{-3}			
		4	0.336	9.84×10^{-5}			
1200	10^{-3}	1	7.40×10^{-2}	2.46×10^{-6}	2.46×10^{-6}	1.000	3.08×10^7
		2	0.427	7.27×10^{-7}	3.18×10^{-6}	1.000	2.38×10^7
		3	0.681	2.41×10^{-7}	3.42×10^{-6}	1.000	2.21×10^7
		4	0.816	8.45×10^{-8}	3.51×10^{-6}	1.000	2.16×10^7
		5	0.886	3.37×10^{-8}	3.54×10^{-6}	1.000	2.13×10^7
		6	0.924	1.53×10^{-8}	3.56×10^{-6}	1.000	2.13×10^7
		7	0.947	7.63×10^{-9}	3.56×10^{-6}	1.000	2.12×10^7
		34	88.46	1.01×10^{-6}			

significantly decrease the intermediate timescale between this set and the lowest eigenvalue.

In general, as a rough guide, about 3–4 orders of magnitude separation in timescale appears nec-

essary between the lowest eigenvalue(s) and the next eigenvalue associated with an eigenvector with appreciable flux overlap before an intermediate steady state can exist.

Table 8

As for table 7 but for a stepladder model ($\alpha = 4$ kcal/mol)

T (K)	P (Pa)	i	λ_i	$r_i f_i / F$	$(S/F)_i$	$(D/F)_i$	$(\omega D/S)_i$
600	10^{-3}	1	2.02×10^{-5}	2.85×10^{-8}	2.85×10^{-8}	1.000	3.75×10^9
		2	3.94×10^{-5}	-1.01×10^{-11}	2.85×10^{-8}	1.000	3.75×10^9
		3	5.91×10^{-5}	-5.39×10^{-12}	2.85×10^{-8}	1.000	3.75×10^9
		4	3.85×10^{-3}	4.46×10^{-11}	2.85×10^{-8}	1.000	3.75×10^9
		5	0.279	3.00×10^{-11}			
600	10^5	1	3.37×10^{-9}	0.99853	0.99853	1.47×10^{-3}	1.57×10^7
		2	3.15×10^{-8}	6.26×10^{-4}	0.99916	8.44×10^{-4}	9.03×10^6
		3	4.60×10^{-8}	1.86×10^{-5}	0.99917	8.25×10^{-4}	8.83×10^6
		4	3.83×10^{-3}	4.11×10^{-8}			
1200	10^{-3}	1	4.19×10^{-2}	-7.29×10^{-13}			
		2	4.65×10^{-2}	2.52×10^{-9}	2.52×10^{-9}	1.000	3.00×10^{10}
		3	5.16×10^{-2}	-6.82×10^{-15}			
		4	5.70×10^{-2}	-3.47×10^{-19}			
		5	0.240	-9.30×10^{-13}	2.51×10^{-9}	1.000	3.01×10^{10}
		6	0.255	1.29×10^{-9}	3.80×10^{-9}	1.000	1.99×10^{10}
		10	0.529	9.76×10^{-10}	4.78×10^{-9}	1.000	1.58×10^{10}
		20	1.16	5.36×10^{-10}	6.05×10^{-9}	1.000	1.25×10^{10}
		28	1.74	1.90×10^{-10}	6.59×10^{-9}	1.000	1.15×10^{10}
		32	1.95	6.00×10^{-11}	6.65×10^{-9}	1.000	1.14×10^{10}
		34	88.6				

5.3. Application of the intermediate timescale method

As discussed in section 5.2, if a system has an eigenvalue spectrum that can be separated into distinct timescales than an intermediate steady state exists. More precisely stated, if the condition (44) holds then eqs. (46) and (47) can be used to predict the intermediate-steady-state values of the D and S fluxes.

A typical application of the intermediate timescale method can be seen in table 7 for an exponential model. At low temperatures, $T = 600$ K, there is an appreciable timescale difference between the first and second eigenvalues so that condition (48) holds and eq. (50) can be applied,

$$S = r_1 f_1,$$

while,

$$D = F - S.$$

At higher temperatures the eigenvalue spectrum becomes more dense and an intermediate timescale may exist only at low pressures. For example at $T = 1200$ K, $P = 10^5$ Pa for the exponential model fig. 3b shows no intermediate steady state which corresponds to a lack of separation in the eigenvalue spectrum as seen in table 4b. However at a lower pressure ($P = 10^{-3}$ Pa) at the same temperature fig. 2b shows an intermediate steady state which is evaluated in table 7 using eq. (47).

Because it is not obvious that an intermediate steady state will exist it is preferable to use the exact treatment as such elevated temperatures.

For a collision model such as the stepladder model there are additional complications. If the stepsize of the collision model α [13] is equal to the grainsize of the calculation then only jumps between nearest neighbour energy levels (or basis functions in EFBS notation [13]) are allowed. Then the lowest eigenvalue will be separated from the higher eigenvalues and eq. (50) can be applied as above to yield the intermediate timescale results. However, if α is say some higher multiple of the grainsize then a family of non-interacting energy jumps between non-nearest neighbour energy levels is possible. This gives rise to a nearly degenerate set of lowest-lying eigenvalues separated from the higher eigenvalues (see table 8). Thus in such cases this set must be included to yield the intermediate timescale result from eq. (47) instead of eq. (50).

5.4. Comparison of the intermediate timescale and sink methods

It is apparent that if the temperature is high enough an intermediate steady state will not exist and an exact treatment is necessary. However, chemical activation studies are generally made at very low temperatures so that there is an appreciable timelag between relaxation of the incoming

Table 9

A comparison of the intermediate-steady-state rate constant $\omega D/S$ calculated by the sink method versus the intermediate timescale (IT) method and the exact method for various collision models at low pressure ($P = 10^{-3}$ Pa). The % deviation is a measure of the deviation of the sink value from the IT or exact values

Model	$T(K)$	Sink	IT	Exact	% deviation
SCA	300	2.24×10^6	2.24×10^6		0.0
EXP($\alpha = 4$)		1.27×10^7	1.30×10^7		-2.3
EXP($\alpha = 2$)		6.79×10^7	7.23×10^7		-6.1
SCA	373	2.27×10^6	2.27×10^6		0.0
EXP($\alpha = 4$)		1.34×10^7	1.40×10^7		-4.3
EXP($\alpha = 2$)		7.34×10^7	8.15×10^7		-9.9
SCA	500	2.31×10^6	2.31×10^6		0.0
EXP($\alpha = 4$)		1.46×10^7	1.59×10^7		-8.0
EXP($\alpha = 2$)		8.14×10^7	9.99×10^7		-18.5
SCA	600	2.33×10^6	2.33×10^6	2.33×10^6	0.0
EXP($\alpha = 4$)		1.55×10^7	1.78×10^7	1.78×10^7	-13
EXP($\alpha = 2$)		8.70×10^7	1.19×10^8	1.19×10^8	-27

Table 10

As for table 9 but at high pressure ($P = 10^5$ Pa)

Model	$T(K)$	Sink	IT	Exact	% deviation
SCA	300	2.24×10^6	2.24×10^6		0.0
EXP($\alpha = 4$)		3.90×10^6	3.92×10^6		-0.5
EXP($\alpha = 2$)		5.94×10^6	5.94×10^6		0.0
SCA	373	2.27×10^6	2.27×10^6		0.0
EXP($\alpha = 4$)		4.59×10^6	4.60×10^6		-0.2
EXP($\alpha = 2$)		7.66×10^6	7.68×10^6		-0.3
SCA	500	2.31×10^6	2.31×10^6		0.0
EXP($\alpha = 4$)		6.96×10^6	6.99×10^6		-0.4
EXP($\alpha = 2$)		1.38×10^7	1.39×10^7		-0.7
SCA	600	2.33×10^6	2.33×10^6	2.33×10^6	0.0
EXP($\alpha = 4$)		1.14×10^7	1.16×10^7	1.16×10^7	-1.7
EXP($\alpha = 2$)		2.55×10^7	2.62×10^7	2.62×10^7	-2.7

flux and the onset of significant thermal reaction. Confining ourselves to low temperatures for which an intermediate timescale exists it is instructive to compare the traditional sink treatment with the intermediate timescale method.

Since the steady-state assumption (9) holds in the intermediate timescale, any discrepancy between the two methods is bound to be due to the remaining assumption used in the sink method, i.e. the assumption of an irreversible sink for stabilization. This assumption is reasonable for strong-collision models as shown in tables 9 and 10 but discrepancies appear for weak-collision models. At low temperatures a strong-collision model will give the impression of irreversible stabilization in one hop. However, weak-collision models with their small energy jumps will have a significant likelihood of activating compared to deactivating events, especially as the energy jumps become smaller. This will result in an overestimate of stabilization by the sink method and hence an underestimate of $\omega D/S$ as shown in tables 9 and 10. This overestimate of stabilization decreases as one goes to high pressure from $P = 10^{-3}$ Pa to $P = 10^5$ Pa. At the higher pressure, deactivation is much more likely because of the higher collision frequency so that the irreversible stabilization assumption is not as bad.

6. Conclusion

We have shown here that the population of

reactant levels as well as decomposition and stabilization fluxes in a chemically activated unimolecular reaction can be obtained exactly in the usual master-equation model in terms of eigenvectors and eigenvalues of the corresponding thermal activation problem. The analysis has been limited here to the case of a steady-state flux of reactants created by chemical activation but it can be extended in a straightforward manner to the case of a time-dependent activation process and to the consideration of the fate of a single molecule. The latter case can be represented as a δ function like pulse of flux.

Our analysis shows that in the case of a steady-state chemical activation process started at $t = 0$ one can generally expect to observe three separate timescales in the reactant level populations and stabilization and decomposition fluxes. An initial transient or incubation period when the reactant population is rapidly building up. An intermediate timescale when the reactant population as a whole grows linearly with t but the population of above barrier levels is essentially constant as are the stabilization and decomposition fluxes. This timescale is of the greatest experimental significance being studied in most earlier work on chemical activation. However, in the long-time limit the system will approach another steady state. If the decomposition is taken to be irreversible then the long-time steady state will show a stationary population of reactants and zero stabilization flux. The reservoir of thermally stabilized molecules has then

grown to the point that the deactivation of chemically activated molecules is precisely balanced by the additional decomposition flux due to thermal activation. Thus one must be careful to distinguish the long-time steady state from any intermediate steady state and the existence of the latter must not be taken for granted. Our analysis shows that the existence of an intermediate steady state is conditional and ensured only if the eigenvalues of the rate matrix of the corresponding thermal reaction problem can be separated into a set of small and a set of much larger eigenvalues. A similar condition arises in the calculation of a well-defined thermal rate coefficient which can be taken to be the smallest eigenvalue only if this is well separated from the rest. Thus we can expect a well-defined intermediate steady state to exist if the unimolecular reaction shows a simple exponential decay for all but an insignificant initial incubation period. This is generally the case at low temperatures but at high temperatures the initial non-exponential decay may become significant. The intermediate steady state may therefore be expected to become shorter with increasing temperature (and thermal rate coefficient) and may essentially disappear. This has been observed in our test calculations.

Our analysis has also led to a simple approximate prediction of the intermediate timescale properties. This intermediate timescale method is an improvement on the usual sink method since we no longer neglect the possibility of collisional activation. This is important particularly in the case of weak dispersive energy transfer where the reactant molecule may well recross the energy surface dividing activated from "sink" configurations. Our test calculations indicate that 10–30% errors may easily arise in the sink method at low pressures.

Finally we have also wanted to illustrate the straightforward extension of the very efficient equilibrium finite-basis (EFBS) method from thermally activated [13] to chemically activated unimolecular reactions. The method is particularly useful in cases where the energy transfer is weak and there is a need to calculate more than one eigenvalue. However, in our present test calculations we have not exploited the possibility of im-

proving efficiency by choosing a non-uniform grid of basis functions but used a uniform grid of 1 kcal/mol steps up to a maximum cutoff at 100 kcal/mol for $T = 1200$ K.

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